Distilling a 9B-Parameter Language Model into a 1.5B Student for Offline Text-to-SQL

SENSEI-SQL: Execution-verified Knowledge Distillation

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Natural Language Processing

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Outline

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3. Method
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Problem: natural language → SQL, but the good models are too big

Text-to-SQL: turn a question into an executable query over a given database.

Example

"Which singers are older than the average age?"

```
SELECT Name, Age FROM singer
WHERE Age > (SELECT AVG(Age) FROM singer)
```

- Lets non-programmers query data directly
- Many databases must stay **on-premise**: data cannot leave the machine

The cost of a strong LLM

Qwen3.5-9B (FP8), our teacher:

- $\approx$ 15.5 GiB GPU memory
- $\approx$ 0.95 s per query on GPU
- $\approx$ 10 s per query on a CPU

Question

Can a **1.5B** model keep most of the accuracy?

Objectives and contributions

1. An **execution-verified distillation pipeline**: a 9B teacher produces 23,703 clean question/SQL pairs over 146 training databases, with a guarantee of no dev-set leakage.
2. A **QLoRA-trained 1.5B student** that retains **98.8%** of the teacher's Spider-dev accuracy, and a deployment path down to a 0.92 GiB GGUF file.
3. **Two ablations** (training-data source, amount of teacher data) and an **error analysis** with a significance test.
4. A fully **offline web application** with a read-only, safe SQL runner.

Benchmark and metric

Spider (Yu et al., 2018)

- Cross-domain: dev databases are *unseen* in training
- Dev set: 1,034 questions, 20 databases
- Hardness levels: easy / medium / hard / extra-hard

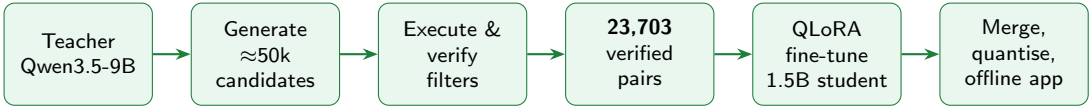
Execution accuracy (EX)

- A prediction is correct if it returns the same result as the gold SQL
- Official *test-suite* scorer (Zhong et al., 2020): several database copies reduce false positives
- Standard error at 1,034 questions: $\approx \pm 1.3$ points

Tools

Teacher: Qwen3.5-9B (FP8) Student: Qwen2.5-Coder-1.5B-Instruct Training: QLoRA (4-bit NF4) Serving: vLLM, llama.cpp

Pipeline overview



- The teacher's signal is **data**: new questions for each training database, plus its own SQL for Spider's training questions.
- Because SQL *executes*, every pair can be **verified** before the student sees it.
- One prompt format, one schema formatter and one safe executor are shared by all stages.

Prompt and safe execution

Schema text

- `CREATE TABLE` statements, keys, plus 3 example rows per table
- One system + one user message; target = bare SQL
- Student is always used *zero-shot*

Model-written SQL is never run unchecked

1. `sqlglot`: exactly one `SELECT`/`WITH` statement
2. Read-only connection (`mode=ro`, `query_only`)
3. Timeout and row cap

Result comparison

Same rows, ignoring row order (unless `ORDER BY`) and column names. The same runner is used for filtering, scoring and the web app.

Teacher data generation

Two sources (training databases only; an assertion forbids dev databases)

1. **Synthesis**: per database, 6 SQL patterns $\times$ 5 runs $\times$ 10 pairs, temperature 0.9 $\Rightarrow$ 41,055 candidates
2. **Relabelling**: teacher answers Spider's 8,659 training questions greedily

Patterns: single-table, aggregation, joins, nested subqueries, order/limit, set operations.

Filter stage	Synth.	Relabel
Candidates	41,055	8,659
Parses	40,797	8,656
Executes	39,041	8,388
Non-empty	33,236	6,839
Verified	18,986	5,410
Deduplicated	**18,346**	**5,357**

Verified $= \geq 3$ of 5 re-samples give the same *result* (synthesis) or match gold (relabel). Total: **23,703** pairs.

Student fine-tuning (QLoRA)

Student	Qwen2.5-Coder-1.5B-Instruct
Base weights	4-bit NF4, bf16 compute
LoRA	rank 16, $\alpha = 32$, all linear layers
Training	2 epochs, lr 2×10^{-4}, cosine
Batch	8×2 (effective 16)
Loss	on the SQL only
Seed	0 (all runs)

Three variants, one recipe

- (a) **human**: Spider train, 8,406 examples
- (b) **teacher**: 23,149 examples
- (c) **combined**: 31,552 examples

Only the data changes. `group_by_length` cut step time from $\approx$12 s to $\approx$5 s.

Experimental setup

- **Benchmark**: Spider dev, 1,034 questions
- **Decoding**: greedy, max 256 tokens, fixed seed, vLLM 0.19.1
- **GPU**: NVIDIA RTX 4000 Ada (20 GB)
- **CPU**: Intel i9-14900K, 32 threads (first 100 dev questions)
- **Latency**: batch size 1 (median, p95)
- **Throughput**: batch size 32
- **Memory**: peak VRAM, fixed 2 GiB KV cache for all models
- **Baselines**: teacher 0-shot / 3-shot, untuned student

Statistical caution

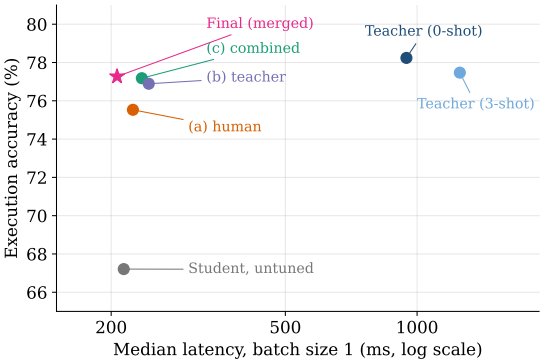
One seed per setting; gaps under $\approx$2.5 points are not established.

Main results on Spider dev

Model	EX	Easy	Med.	Hard	Extra	p50 (ms)	q/s	VRAM	Size
Teacher Qwen3.5-9B, 0-shot	**.782**	.940	.834	.707	.488	945	4.2	15.5	13.1
Teacher, 3-shot	.775	.944	.848	.667	.440	1252	1.7	16.0	13.1
Student, no fine-tuning	.672	.867	.729	.511	.398	214	42.7	6.5	2.9
(a) + human data	.755	.907	.812	.632	.506	224	38.7	7.2	3.0
(b) + teacher data	.769	.911	.834	.713	.440	244	42.4	7.2	3.0
(c) + human + teacher	.772	.923	.807	.707	.518	235	39.0	7.2	3.0
(c) merged fp16 (final)	**.773**	.919	.812	.713	.512	**206**	**44.1**	**6.5**	**2.9**

VRAM and size in GiB. Adapter rows (a)–(c) run through vLLM, which adds latency and memory; the merged model removes that overhead.

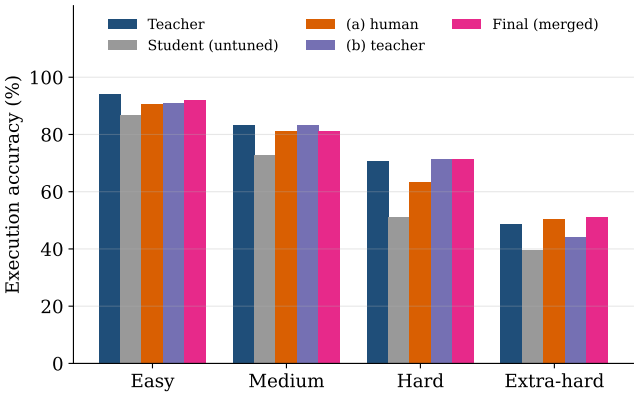
The student keeps 98.8% of the teacher's accuracy



Execution accuracy	**98.8%** retained
Latency (batch 1)	4.6× faster
Throughput (batch 32)	10.6×
GPU memory	2.4× less
Disk size	4.5× smaller

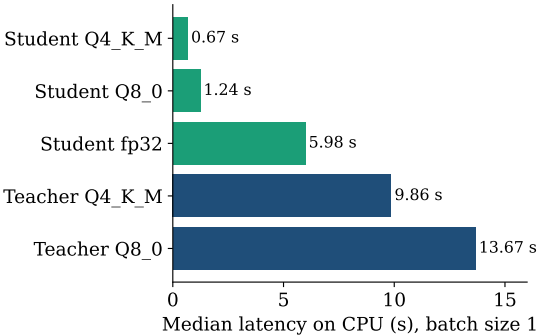
77.3% vs 78.2% on 1,034 questions. Distillation closed 92% of the gap between the untuned student and the teacher.

Where the gains come from: accuracy by hardness



- Hard: 0.51 → 0.71
- Extra-hard: 0.40 → 0.51
- Easy: already 0.87 → 0.92
- Extra-hard matches the teacher (0.512 vs 0.488, within noise)

CPU deployment: a 0.92 GiB model that answers in under a second



Model	Format	Size	EX
Student	fp32	2.89	.83
Student	Q8_0	1.53	.80
Student	Q4_K_M	**0.92**	.77
Teacher	Q8_0	9.11	.84
Teacher	Q4_K_M	5.38	.84

14.6× faster than the teacher at Q4_K_M (11× at Q8_0).

Caveat: 100 easier-than-average questions (±4 points); the accuracy gap here is larger and may partly be quantisation loss.

Ablation 1: which training data?

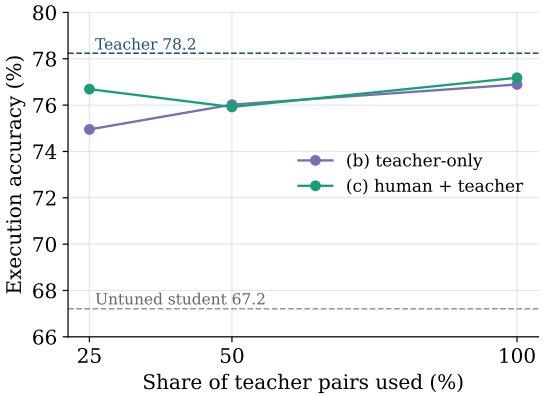
Training data	EX	Easy	Med.	Hard	Extra
None (untuned)	.672	.867	.729	.511	.398
(a) Human only	.755	.907	.812	.632	.506
(b) Teacher only	.769	.911	.834	.713	.440
(c) Human + teacher	**.772**	.923	.807	.707	.518

- Every variant helps: +8 to +10 points
- Teacher data is *at least as good as* human labels (inside the noise)

Why combine both?

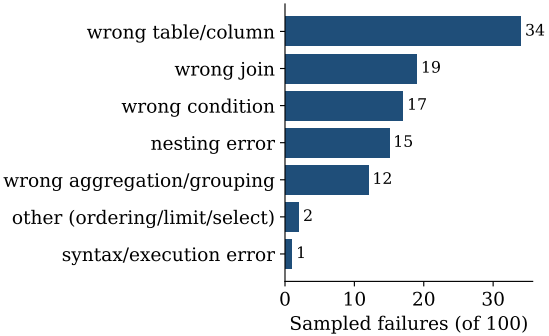
Teacher data helps most on **hard** questions (0.713 vs 0.632); human data helps most on **extra-hard** (0.506 vs 0.440). The combination gets both.

Ablation 2: how much teacher data?



- Teacher-only: about **+1 point per doubling**, no sign of saturation
- Combined: flat within noise (0.759–0.772)
- Human data + **25%** of the teacher pairs (≈5.8k) already reaches 0.767
- The dip at 50% is about one standard error: not a real effect

What the student gets wrong



236 errors on dev; 100 sampled and categorised.

- Only **1%** do not execute: the student writes *valid* SQL that answers a slightly different question
- **Schema linking** dominates: wrong table/column 34%, wrong join 19%
- Some "failures" are benchmark artefacts (case, ties, ambiguous questions)

Student vs. teacher, question by question

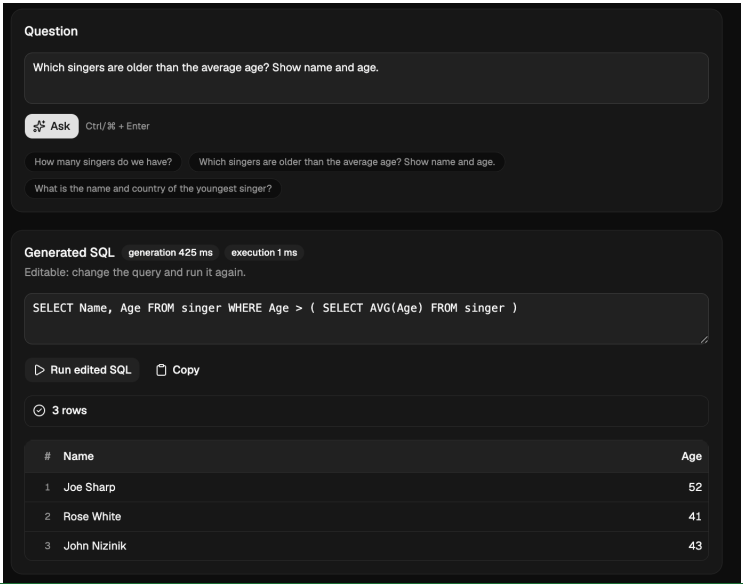
Outcome	Questions
Both correct	706
Student only	92
Teacher only	103
Both wrong	133

- The two models fail on largely **different** questions
- The teacher wins 11 more disagreements (1.1 points)
- **Exact McNemar test**, 195 disagreements: $p = 0.47$

Conclusion

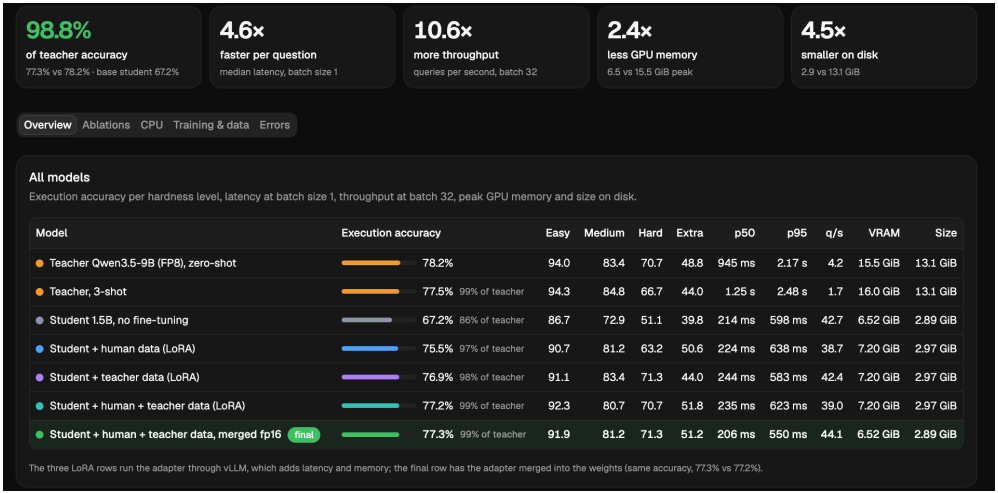
There is no statistical evidence that the student is worse than the teacher on Spider dev.

Offline web application: live demo



- FastAPI back end + React front end
- 166 Spider databases or an **uploaded SQLite file**
- Suggested questions per database
- Editable SQL, result table, timings, history
- Back ends: vLLM, transformers, llama-server, Ollama
- Read-only `SELECT`; one self-correction retry

Results dashboard (built into the app)



Every table and chart in this talk is also in the app's *Results* page (tabs: Overview, Ablations, CPU, Training & data, Errors), with the caveats shown next to the numbers.

Limitations

- **One benchmark**: Spider dev (SQLite); BIRD and larger, messier schemas were not tested
- **Single seed**: gaps below $\approx$2.5 points (e.g. teacher-only vs. human-only) are not established
- **Execution accuracy** can credit lucky matches and penalise ambiguous questions
- **The teacher is not gold**: filters check executability and consistency, not truth
- **CPU numbers**: 100 questions, one machine; quantisation loss is not separated from noise
- **Sequence-level distillation**: we distil through data, not by matching logits

Conclusion and future work

Conclusion

A 1.5B student trained on execution-verified teacher data reaches **77.3%** execution accuracy, **98.8%** of the 9B teacher's 78.2%, with 4.6× lower latency, 10.6× higher throughput and 4.5× smaller size.

At 4 bits it is **0.92 GiB** and answers in under a second on a CPU: a private, offline text-to-SQL assistant.

Future work

- Multiple seeds, mean ± std
- Full-dev evaluation of the GGUF models
- More teacher data (curve not saturated)
- Better schema linking
- Evaluate on BIRD

Thank you

Questions?

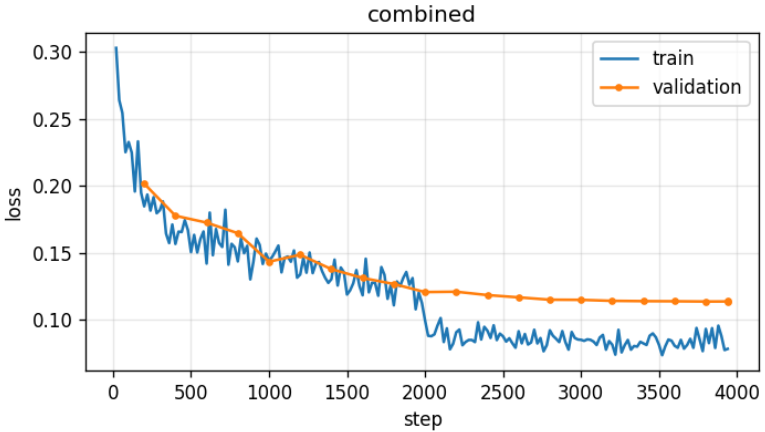
Code, report and results are in the project repository.

Backup: training runs

Run	Examples	Steps	Minutes	Best val. loss
Human	8,406	1,052	94	0.065
Teacher	23,149	2,894	250	0.106
Combined	31,552	3,944	350	0.114
Teacher 25% / 50%	5,786 / 11,566	724 / 1,446	64 / 125	0.131 / 0.119
Combined, 25% / 50% teacher	14,185 / 19,973	1,774 / 2,498	161 / 225	0.136 / 0.123

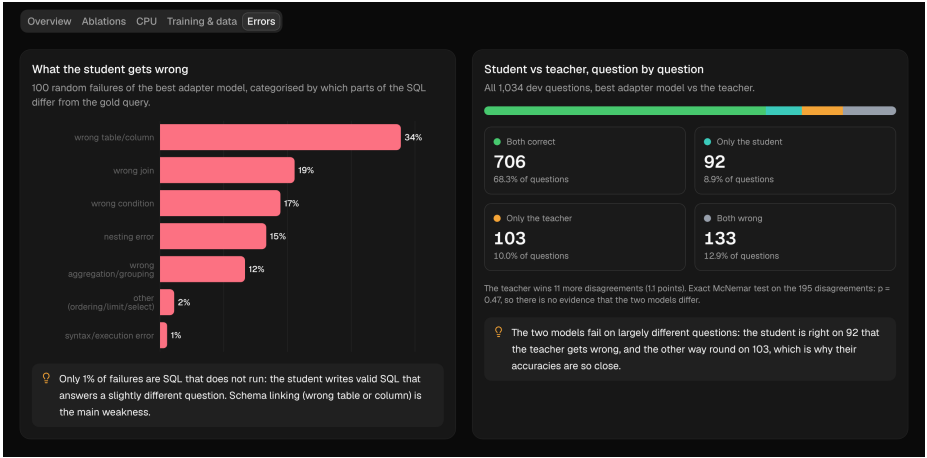
Validation losses are measured on each variant's own held-out data and are *not* comparable across variants.
Total training $\approx$ 21 h on one RTX 4000 Ada.

Backup: training curve



Combined run (31,552 examples, 3,944 steps): validation loss flattens after ≈2,500 steps; the lowest-validation checkpoint is kept.

Backup: error analysis in the app



Backup: reproducing the pipeline

```
make install && make download # environment, Spider, models, scorer
make check && make test # setup check, unit tests
make teacher # teacher data (~7.5 h)
python -m sensei_sql.train configs/train/combined.yaml
python -m sensei_sql.train.export merge --adapter models/adapters/combined
python -m sensei_sql.eval --model models/student-merged
python -m sensei_sql.train.export gguf
make results && make app # tables/plots, offline web app
```

Pinned: Python 3.10.12, torch 2.10.0+cu128, vLLM 0.19.1, transformers 5.20.2, TRL 1.14.2, PEFT 0.21.2.